

$$g = \frac{1}{q(\mu_n \cdot n + \mu_p \cdot p)} \downarrow = \frac{1}{q(\mu_n + \mu_p) n_i}$$

Si es intrínseco

Error, la movilidad sí cambia con $T \rightarrow *$

$$n_i = c \cdot T^{3/2} e^{-\frac{E_g}{2kT}} \Rightarrow g = \frac{1}{c \cdot q(\mu_n + \mu_p)} (T)^{-3/2} e^{\frac{E_g}{2kT}} = c' (T)^{-3/2} e^{\frac{E_g}{2kT}}$$

$$R(T) = g(T) \cdot \frac{\text{Largo}}{\text{Area}} = c'' (T)^{-3/2} e^{\frac{E_g}{2kT}}$$

$$R_{298} = R(298) \quad \frac{R(T)}{R_{298}} = \frac{c'' (T)^{-3/2} e^{\frac{E_g}{2kT}}}{c'' (298)^{-3/2} e^{\frac{E_g}{2k \cdot 298}}} =$$

$$R(T) = R_{298} \left(\frac{T}{298} \right)^{-3/2} \exp \left\{ \frac{E_g}{2K} \left(\frac{1}{T} - \frac{1}{298} \right) \right\}$$

$$R(T) = R_{298} \left(\frac{T}{298} \right)^{-3/2} \exp \left\{ - \frac{E_g}{2K} \left(\frac{1}{298} - \frac{1}{T} \right) \right\}$$

$$\left[\text{Si } \frac{T}{298} \sim 1 \right] \Rightarrow R(T) = R_{298} \exp \left\{ - \frac{E_g}{2K} \left(\frac{1}{298} - \frac{1}{T} \right) \right\}$$

Elige β

$$\ln(R(T)) = \ln R_{298} - \beta \left(\frac{1}{298} - \frac{1}{T} \right) = a + \beta \cdot x$$

→ ajustat la
recta per python

$$\beta \left(\frac{1}{298} - \frac{1}{T} \right) = \ln \frac{R_{298}}{R(T)} \Rightarrow \beta = \left(\frac{1}{298} - \frac{1}{T} \right)^{-1} \ln \frac{R_{298}}{R(T)}$$

R_{258} lo sacamos de la simulación

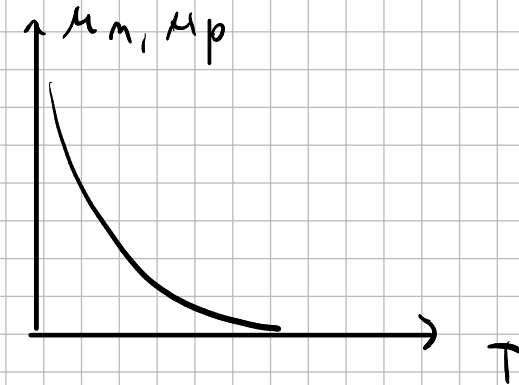
$$R_{258} = \frac{553 \text{ mV}}{55,9 \mu\text{A}} = 3832 \Omega$$

¿Posible SC? $\rightarrow$ del β despejar E_g . $E_g = \beta \cdot 2k$

⊗ $\rightarrow$ La movilidad depende de 2 cosas: las interacciones con la red cristalina y las interacciones eléctricas con los átomos de impurezas ionizados. El termistor tiene que estar hecho con un material intrínseco (+ temperatura $\Rightarrow$ -resistencia, tiene que ser intrínseco), así que no hay impurezas ionizadas. En tal caso, las movilidades dependen de la temperatura de la siguiente manera:

$$\mu_n = k_n \cdot (T)^{-3/2}, \quad \mu_p = k_p \cdot (T)^{-3/2}$$

$\uparrow$ $\uparrow$
constantes arbitrarias



$$\Rightarrow g = \frac{1}{q(\mu_n + \mu_p)n_i}$$

$$n_i \propto T^{3/2} \exp\left\{-\frac{E_g}{2kT}\right\}$$
$$\mu_n + \mu_p = (k_n + k_p) T^{-3/2}$$

$$\Rightarrow g = \frac{1}{q} \frac{1}{k_n + k_p} \cdot \frac{1}{T^{-3/2} \cdot T^{3/2}} \exp\left\{ \frac{E_g}{2kT} \right\}$$

$$\Rightarrow g \propto \exp\left\{ \frac{E_g}{2kT} \right\}$$

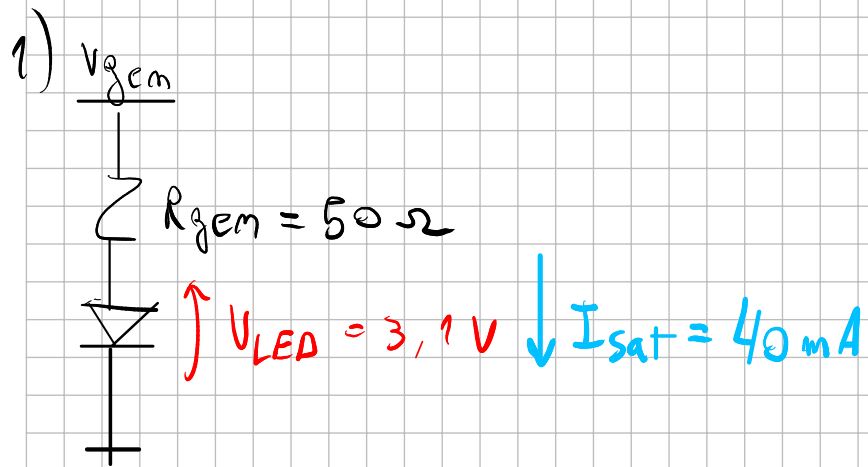
$$\Rightarrow R(T) = g \cdot \frac{L}{A} = C \cdot \exp\left\{ \frac{E_g}{2kT} \right\}$$

$$R_{298} = C \cdot \exp\left\{ \frac{E_g}{2k(298)} \right\}$$

$$R(T)/R_{298} = \exp\left\{ \frac{E_g}{2k} \left(\frac{1}{T} - \frac{1}{298} \right) \right\} \quad \beta$$

En resumen: no hacía falta aproximar nada

$$\Rightarrow R(T) = R_{298} \exp\left\{ - \left[\frac{E_g}{2k} \right] \left(\frac{1}{298} - \frac{1}{T} \right) \right\}$$



$$V_{gem} = V_{LED(on)} + R_{gem} \cdot I_{LED}$$

$$V_{gem} = 5,1 V \text{ (con el LED saturado)}$$

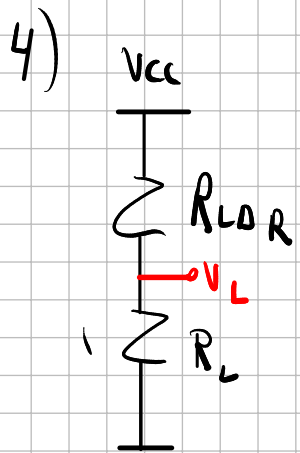
Hacer los gráficos. ¿Se ve algo?

2) obtener $R_{LDR(osc)}$ (DDT)

$$3) R_{LDR(osc)} = \int \frac{L}{A} \Rightarrow \frac{L}{A} = \frac{R_{LDR(osc)}}{\int} \quad \left(T = 300K \text{ según consigna} \right)$$

$$\int = \frac{1}{q(n + p) m_i} \quad m_i = cte (T)^{3/2} e^{-\frac{E_g}{2kT}} \quad \left(\text{¿o usar las masas efectivas?} \right)$$

¿No está dopada? $\rightarrow$ EL LDR es intrínseco, está en la consigna



$$R_{LDR(osc)} = \int_{osc} \cdot \frac{L}{A} \rightarrow \int_{osc} = \frac{A}{L} \cdot R_{LDR(osc)}$$

$$R_{LDR(on)} = \int_{Luz} \cdot \frac{L}{A} \rightarrow \int_{Luz} = \frac{A}{L} \cdot R_{LDR(Luz)}$$

$$V_L = \frac{R_L}{R_L + R_{LDR}} \cdot V_{CC} = \frac{1}{1 + \frac{R_{LDR}}{R_L}} \cdot V_{CC} \Rightarrow \frac{R_{LDR}}{R_L} = \frac{V_{CC} - V_L}{V_L}$$

$$\Rightarrow R_{LDR} = \frac{V_{CC} - V_L}{V_L} \cdot R_L$$

$$\sigma(\text{osc}) = g_{\text{osc}}^{-1} = \frac{L}{A} \cdot \frac{1}{R_{\text{LDR}}(\text{osc})}$$

$$\sigma(\text{LVZ}) = g_{\text{LVZ}}^{-1} = \frac{L}{A} \cdot \frac{1}{R_{\text{LDR}}(\text{LVZ})}$$

$$\Delta\sigma = \sigma(\text{LVZ}) - \sigma(\text{osc}) = \frac{L}{A} \left(\frac{1}{R_{\text{LDR}}(\text{LVZ})} - \frac{1}{R_{\text{LDR}}(\text{osc})} \right)$$

$$R_{\text{LDR}}(\text{LVZ}) = \frac{V_{\text{CC}} - V_{\text{L}}(\text{LVZ})}{V_{\text{L}}(\text{LVZ})} \cdot R_{\text{L}} ; R_{\text{LDR}}(\text{osc}) = \frac{V_{\text{CC}} - V_{\text{L}}(\text{osc})}{V_{\text{L}}(\text{osc})} \cdot R_{\text{L}}$$

$$\Rightarrow \Delta\sigma = \frac{L}{A} \left[\frac{V_{\text{L}}(\text{LVZ})}{V_{\text{CC}} - V_{\text{L}}(\text{LVZ})} \frac{1}{R_{\text{L}}} - \frac{1}{R_{\text{LDR}}(\text{osc})} \right]$$

5) Buscar en el Libro/Clase

$$6) \delta m(t) = \textcircled{cte} e^{-t/\textcircled{\tau}}$$

Para las distintas intensidades tiene que dar lo mismo, no depende de δm .

$$\sigma = q(\mu_n + \mu_p) m_i = q(\mu_n + \mu_p) (m_{i0} + \delta m(t))$$

μ exp

$$\sigma = \sigma_f + (\sigma_i - \sigma_f) e^{-t/\tau}$$

$$\sigma(t) = q \mu m_{i0} + q \mu \delta m_0 e^{-t/\tau} =$$

$$= q \mu m_{i0} + [q \mu m_{i0} + q \mu \delta m_0 - q \mu m_{i0}] e^{-t/\tau} =$$

$$= \underbrace{q \mu m_{i0}}_{\sigma_f} + \underbrace{[q \mu (m_{i0} + \delta m_0) - q \mu m_{i0}]}_{\sigma_i - \sigma_f} e^{-t/\tau}$$

$$\sigma = \alpha + \beta e^{-t/\tau}$$

$$\alpha = q \mu m_{i0}, \quad \beta = \Delta \sigma = q \mu \delta m_0$$

$$R_{LDR} = \frac{L}{A} \frac{1}{\sigma} = \frac{L/A}{\sigma_f + (\sigma_i - \sigma_r) e^{-t/\tau}} = \frac{L/A}{\alpha + \beta e^{-t/\tau}} = \frac{1}{\alpha' + \beta' e^{-t/\tau}}$$

$$(R_{LDR})^{-1} = G_{LDR} = \alpha' + \beta' e^{-t/\tau}$$

$$\alpha' = \alpha \cdot \frac{A}{L}$$

$$\frac{V_L}{V_{CC}} = \frac{R_L}{R_L + R_{LDR}} = R_L \frac{1}{R_{LDR}} \cdot \frac{1}{1 + R_L \cdot \frac{1}{R_{LDR}}} = R_L G_{LDR} \cdot \frac{1}{1 + R_L G_{LDR}} =$$

$$= \frac{(\alpha'' + \beta'' e^{-t/\tau})}{1 + (\alpha'' + \beta'' e^{-t/\tau})}$$

$$\alpha'' = R_L \alpha' = R_L \frac{A}{L} \alpha =$$

$$= R_L \frac{A}{L} \sigma_f = R_L / R_{LDR}(\text{osc})$$

$$\beta'' = R_L \cdot \beta' = R_L \frac{A}{L} \beta = R_L \frac{A}{L} \Delta \sigma = R_L \cdot \Delta G_{LDR}$$



7) ¿? $\rightarrow$ Ecuación ambipolar.

$$\frac{dm(t)}{dt} = \alpha_r [m_i^2 - m(t) \cdot p(t)]$$

$$m(t) = m_0 + \delta m(t) = m_i + \delta m(t)$$

$$p(t) = m_i + \delta m(t)$$

$$\frac{d\delta m(t)}{dt} = \alpha_r [\cancel{m_i^2} - \cancel{m_i^2} - 2m_i \delta m(t) - \delta m(t)^2] = -\alpha_r [2m_i - \cancel{\delta m(t)}] \cdot \delta m(t)$$

Baja inyección: $m_i \gg \delta m(t) \Rightarrow \frac{d\delta m(t)}{dt} = -\alpha_r 2m_i \delta m(t)$

$$\frac{d\delta_M(t)}{dt} = -\alpha_f 2M_i \delta_M(t) \Rightarrow \delta_M(t) = \delta_M(0) e^{-2\alpha_f M_i \cdot t} = \delta_M(0) e^{-t/\tau_M}$$

$$\text{com } \tau_M = \frac{1}{2\alpha_f \cdot M_i}$$

$$\frac{dM}{dt} = -\frac{dF_M}{dx} + g_M - R_M = -\frac{dF_M}{dx} + g_M - \frac{M}{\tau_M}$$

$$J_M = q \mu_M M \mathcal{E} + q D_M \frac{\partial M}{\partial x} = q \mu_M \cdot M \mathcal{E} + q V_{Th} \mu_M \frac{\partial M}{\partial x} \quad \frac{D_M}{\mu_M} = V_{Th}$$

$$\frac{J_M}{(-q)} = -\mu_M \cdot M \cdot \mathcal{E} - V_{Th} \mu_M \partial M / \partial x = F_M \Rightarrow \partial F_M / \partial x = -\mu_M \frac{\partial (M \mathcal{E})}{\partial x} - V_{Th} \mu_M \partial^2 M / \partial x^2$$

$$J_p / q = \mu_p \cdot p \cdot \mathcal{E} - V_{Th} \mu_p \partial p / \partial x = F_p$$

$$\Rightarrow \frac{\partial n}{\partial t} = v_{th} \mu_n \frac{\partial^2 n}{\partial x^2} + \mu_n \frac{\partial(nE)}{\partial x} + (g_n - R_n)$$

$$\Rightarrow \frac{\partial n}{\partial t} = D_n \frac{\partial^2 n}{\partial x^2} + \mu_n \left(E \cdot \frac{\partial n}{\partial x} + n \frac{\partial E}{\partial x} \right) + (g_n - R_n) \rightarrow \text{EC de dif}$$

vale siempre

Condición de neutralidad (CNC): $\delta n = \delta p \quad \forall x$ (Ver. Neamen, cap. 6

p. 17 del Pdf, p. 202 del libro)

$$D' \frac{\partial^2(\delta n)}{\partial x^2} + \mu' E \frac{\partial(\delta n)}{\partial x} + g - R = \frac{\partial(\delta n)}{\partial t}$$

$$\mu' = \frac{\mu_n \mu_p (p - n)}{\mu_n n + \mu_p p}$$

$$D' = \frac{D_n D_p (n + p)}{D_n n + D_p p}$$

Es importante que la temperatura sea cte para que ni no varíe con la temperatura

Ahora: SC infinites $\rightarrow \underline{n = n_i + \delta n(t)} \quad \underline{p = n_i + \delta n(t)} \quad \underline{n = p}$

$$\Rightarrow \underline{\mu' = 0}, \quad D' = \frac{D_n D_p 2n}{(D_p + D_n) n} = 2 \cdot \frac{D_n \cdot D_p}{D_p + D_n}$$

$$D' \cdot \frac{\partial^2 \delta n}{\partial x^2} + (g - R) = \frac{\partial \delta n}{\partial t}$$

$$\text{com } D' = 2 \frac{D_n D_p}{D_n + D_p}$$

Tiempo de vida

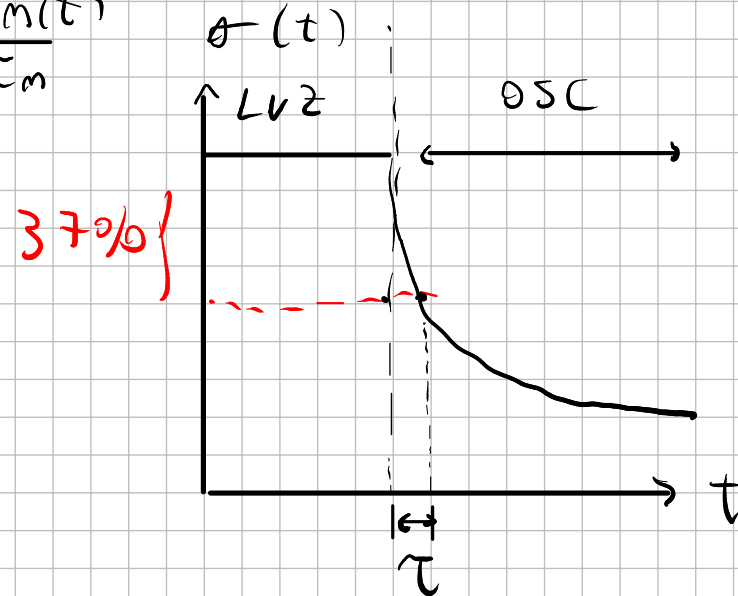
Si la iluminación es uniforme, no hay variaciones espaciales

$g(t) = A \cdot \delta(t) \rightarrow$ impulso en $t = 0$ s

Para $t > 0$ s: $\frac{\partial \delta n}{\partial t} = -R = -\frac{\delta n(t)}{\tau_n}$

$$\Rightarrow \delta n(t) = B e^{-t/\tau} \propto \sigma$$

$$\sigma = q(\mu_n + \mu_p)(n + \delta n(t))$$



"Corroborar la no dependencia de tau con la concentración de portadores en exceso"

--> Para todas las curvas tiene que dar igual

$$\frac{\partial \delta n}{\partial t} = g - \frac{\delta n(t)}{\tau} \quad \text{En estado estacionario: } \partial \delta n / \partial t = 0$$

$$g = \frac{\delta n_{est}}{\tau} \quad \text{Similar a Neamen, ejemplos 6.2 y 6.3}$$

$$\sigma_{(est)} = q(\mu_n + \mu_p)(n_i + \delta n_{(est)}) \rightarrow \text{Estado estacionario, para cuando el LED está iluminado.}$$

$$\delta n_{(est)} = \frac{\sigma_{(est)}}{\mu_n + \mu_p} - n_i = \delta p_{(est)}$$

$$\frac{\partial \delta p}{\partial t} = D_p \frac{\partial^2 \delta p}{\partial x^2} - \mu_p \left(\mathcal{E} \frac{\partial \delta p}{\partial x} + p \frac{\partial \mathcal{E}}{\partial x} \right) + g_p - \frac{p}{\tau_{pt}}$$

$$\frac{\partial \delta n}{\partial t} = D_n \frac{\partial^2 \delta n}{\partial x^2} + \mu_n \left(\mathcal{E} \frac{\partial \delta n}{\partial x} + n \frac{\partial \mathcal{E}}{\partial x} \right) + g_n - \frac{n}{\tau_{nt}}$$

→ ecuaciones de dif dependiente del tiempo para sc homogéneas

$$\text{Si } n = p = n_i \quad ; \quad \delta n = \delta p$$

$$\frac{\partial \delta p}{\partial t} = D_p \frac{\partial^2 \delta p}{\partial x^2} - \mu_p \left(\mathcal{E} \frac{\partial \delta p}{\partial x} + n_i \frac{\partial \mathcal{E}}{\partial x} \right) + g_p - \frac{n_i}{\tau}$$

Para que los e y h se muevan juntos, debe haber un Eint, si no no hay transporte ambipolar
Pero también queremos asumir que los e y h se generan de a partes

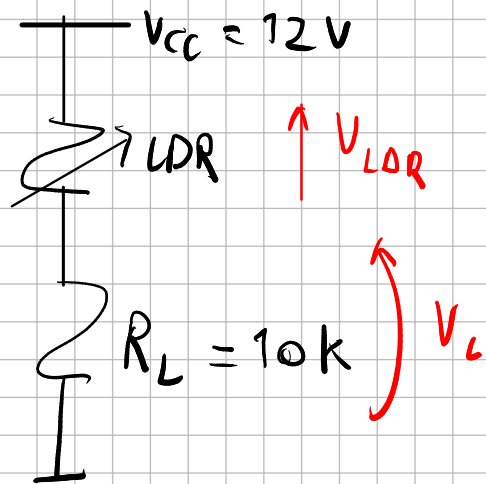
vale siempre que $n_0 \sim p_0$

$$p_0 - n_0 + N_d - N_a = 0 \Rightarrow p_0 = n_0 + N_a - N_d$$

$$\Rightarrow N_a - N_d \ll n_0, p_0 = n_i \quad \text{¿Estará bien?}$$

$$\frac{\partial \delta n}{\partial t} = - \frac{\delta n(t)}{\tau} \quad t > \tau_0$$

¿o tenga mediciones de V_L



$$n \propto \delta n \propto \sigma \propto \frac{1}{g}$$

$$\delta n \propto \Delta \sigma \propto e^{-t/\tau} \Rightarrow \sigma = \sigma_0 + \Delta \sigma = \sigma_0 + K e^{-t/\tau}$$

$$g = \frac{1}{\sigma_0 + K e^{-t/\tau}} \Rightarrow R_{LDR} = \frac{A}{\sigma_0 + K e^{-t/\tau}}$$

$$V_L = \frac{R_L}{R_L + R_{LDR}} \cdot 12V = \frac{12}{1 + \frac{B}{\sigma_0 + K e^{-t/\tau}}} = \frac{12(\sigma_0 + K e^{-t/\tau})}{c + K e^{-t/\tau}}$$

¿ V_L tiene la misma cte de tiempo que δn ?

R_{LDR} como función de V_{CC} y V_L ?

$$V_L = \frac{R_L}{R_L + R_{LDR}} \cdot V_{CC} \Rightarrow R_L + R_{LDR} = \frac{V_{CC}}{V_L} \cdot R_L$$

$$\Rightarrow R_{LDR} = \left(\frac{V_{CC}}{V_L} - 1 \right) R_L \rightarrow \text{Lo grafiqué y no parece exponencial}$$

no importa, el ajuste da bien